

Design, Types and Selection of Tunnel Support Systems

TUNNEL AND CAVERN SUPPORTS

The term supports/reinforcements describe procedures and materials used to improve the stability by maintain the load carrying capacity of rock near the boundaries of underground excavations.

TUNNEL AND CAVERN SUPPORTS

The approaches to this goal are either to reinforce the rockmass or to provide the surrounding rocks with a support resistance with a magnitude being described as the support intensity (external stabilisation).

TUNNEL AND CAVERN SUPPORTS

Rock reinforcement (internal stabilisation) enables the rock strata to support itself, that is, to utilize the inherent strength of the rockmass as much as possible. The purpose of the support is thus to increase the competency of roof strata to achieve full stability after an opening has been made.

TUNNEL AND CAVERN SUPPORTS

When a tunnel and cavern is excavated in all but the most competent of ground conditions it is an inevitable consequence that some form of support will be required if the tunnel is to retain adequate stability and/or maintain sufficient dimensions to facilitate its use in the intended manner.

The form and function of the support will vary according to a wide range of factors apart from just geotechnical considerations and it would seem that for every different tunnel there is a different lining solution.

TUNNEL AND CAVERN SUPPORTS

Tunnel/ cavern supports are grouped into three main forms some or all of which may be used in the construction of a tunnel:

- Temporary ground support
- Permanent ground support
- Secondary lining

During the excavation, at the working face, temporary ground supports are installed to ensure the safe working conditions.

The permanent supports are generally installed after the excavation, behind the working face. The temporary supports usually form a part of the total support requirement.

TUNNEL AND CAVERN SUPPORTS

There are a number of standard types of supports suited to tunnelling applications.

- Rock reinforcement
- Concrete linings
- Shotcrete linings
- Cast iron and steel supports
- Rolled steel joist supports (RSJ)
- Grouting
- Steel mesh
- Other forms of steel supports and lagging
- Spiles or forepoling

Rock Reinforcement

The concept of rock reinforcement or active support in rock is aimed at limiting rock mass deformations by the use of a support medium placed and anchored internally in the rock mass.

Such action is intended to improve the mechanical properties of the rock mass by interlocking blocks of ground or applying additional confining stresses, which are both cases, aim to limit the magnitude of the rock mass deformations.

Rock Reinforcement

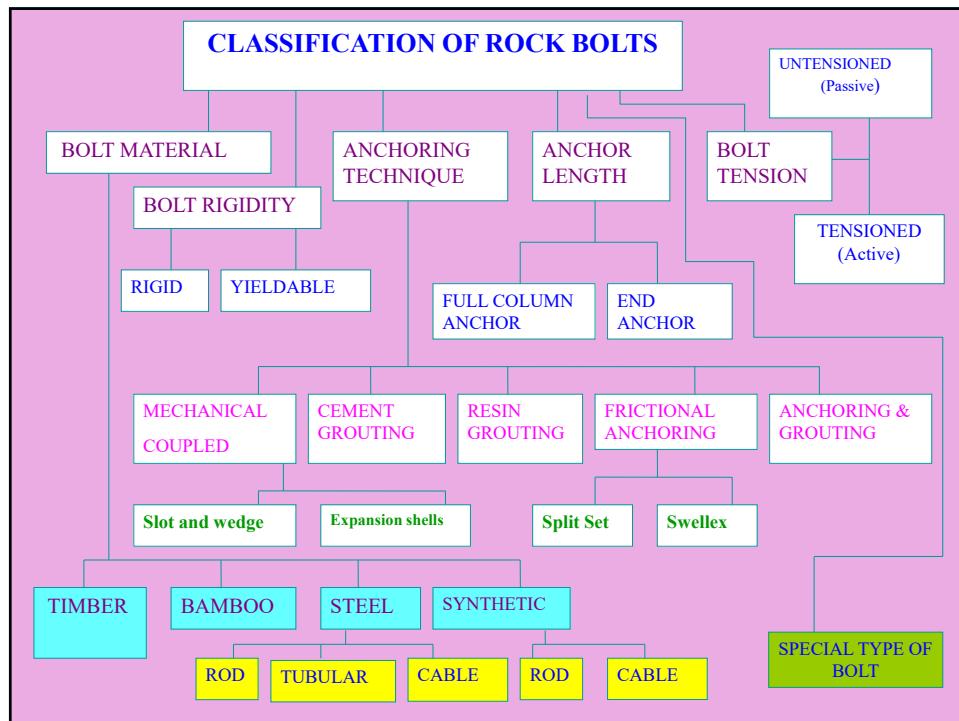
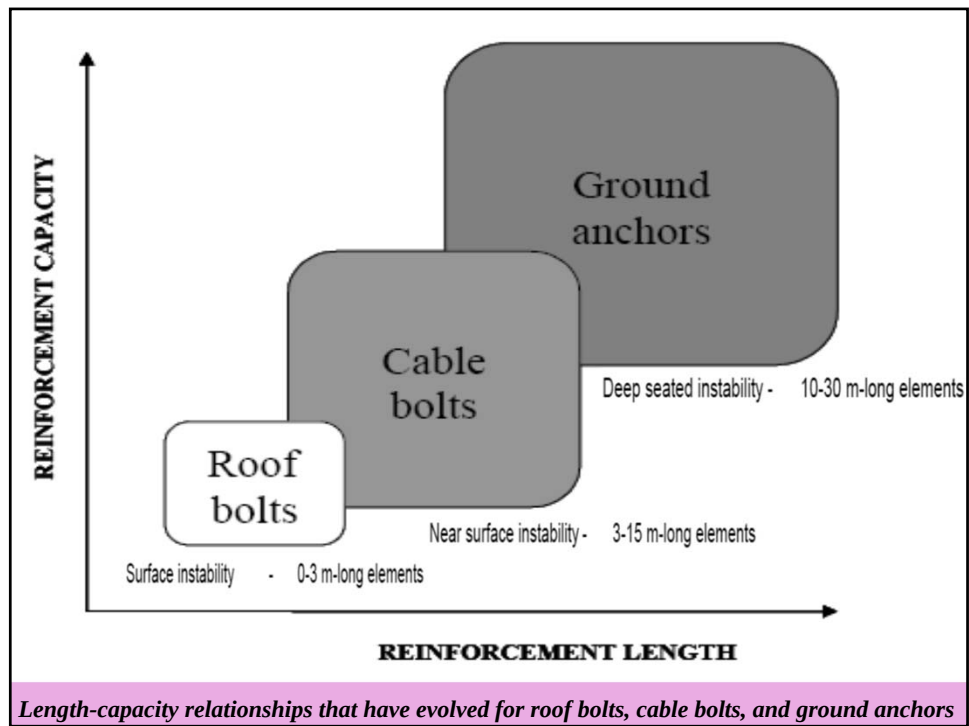
- The increased inter-joint friction increase shear strength and stiffness of rock around excavation. Thereby augments the carrying load carrying capacity of rock.
- The various forms of rock bolts, cable bolts and anchors which are used for rock reinforcement.

Types of roof bolts

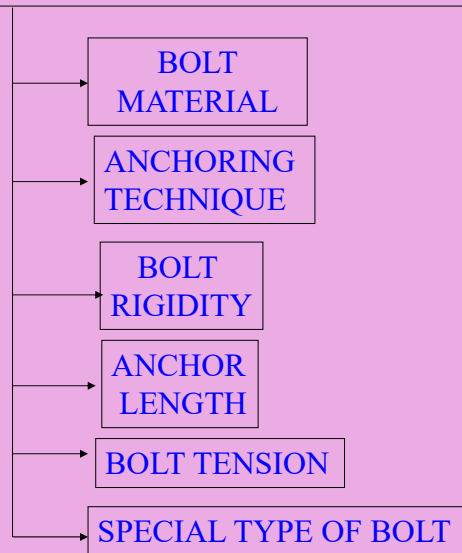
Rock bolting reinforcement practice may be subdivided into three main techniques based on its length and capacity:

1. Roof bolting
2. Cable bolting and
3. Ground anchoring.

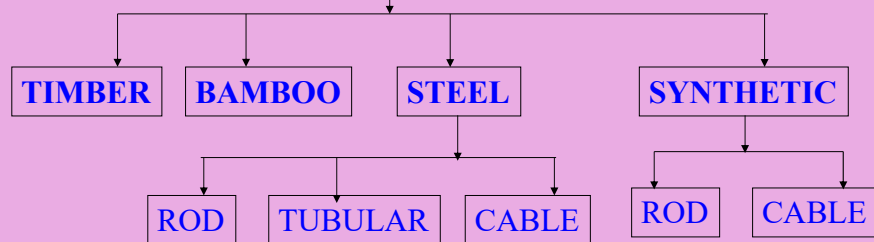
The reinforcement and length–capacity relationship for the three reinforcement techniques are shown in Figure.



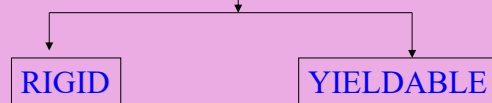
CLASSIFICATION OF ROCK BOLTS

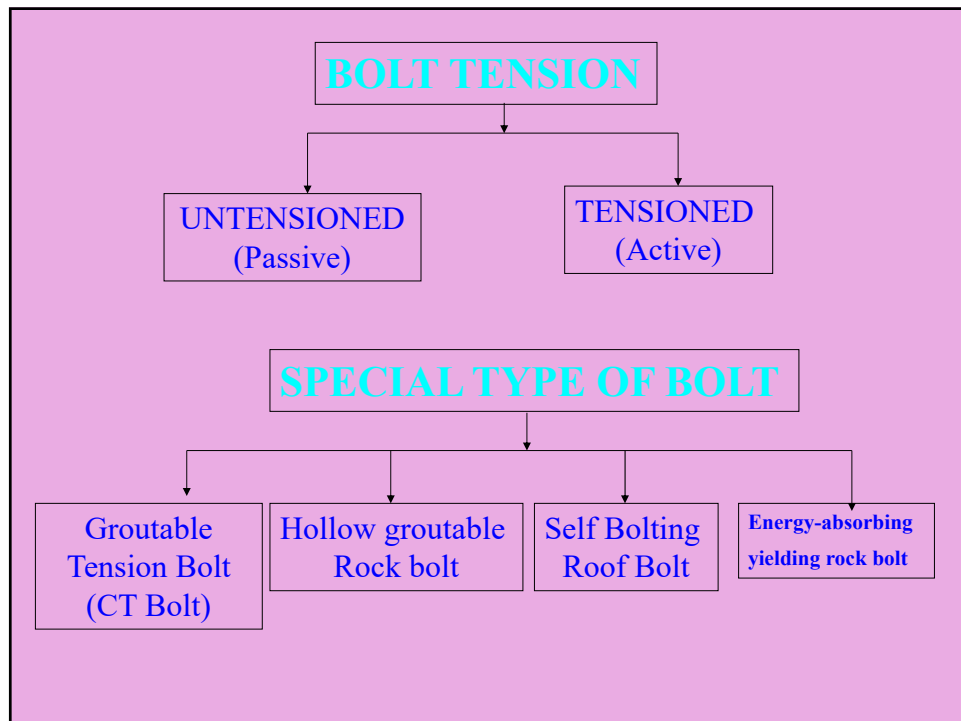
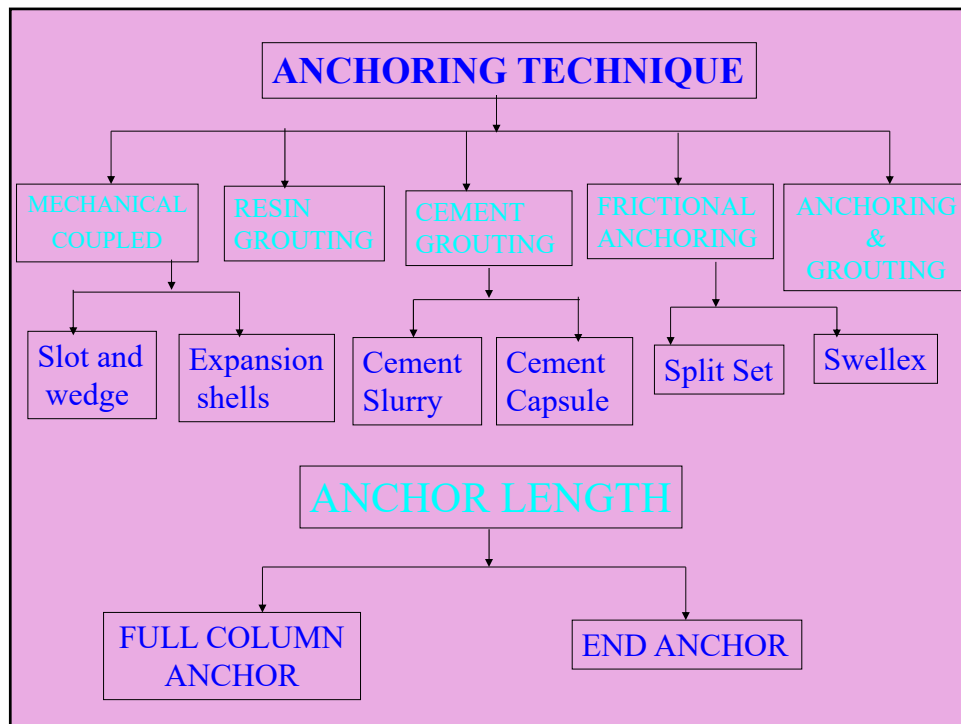


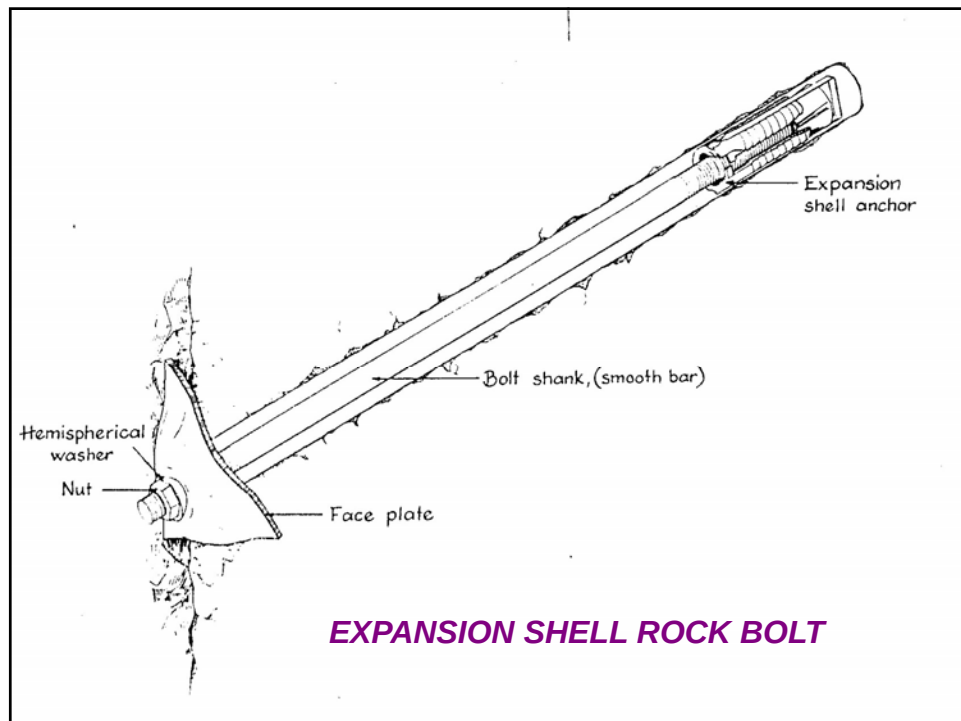
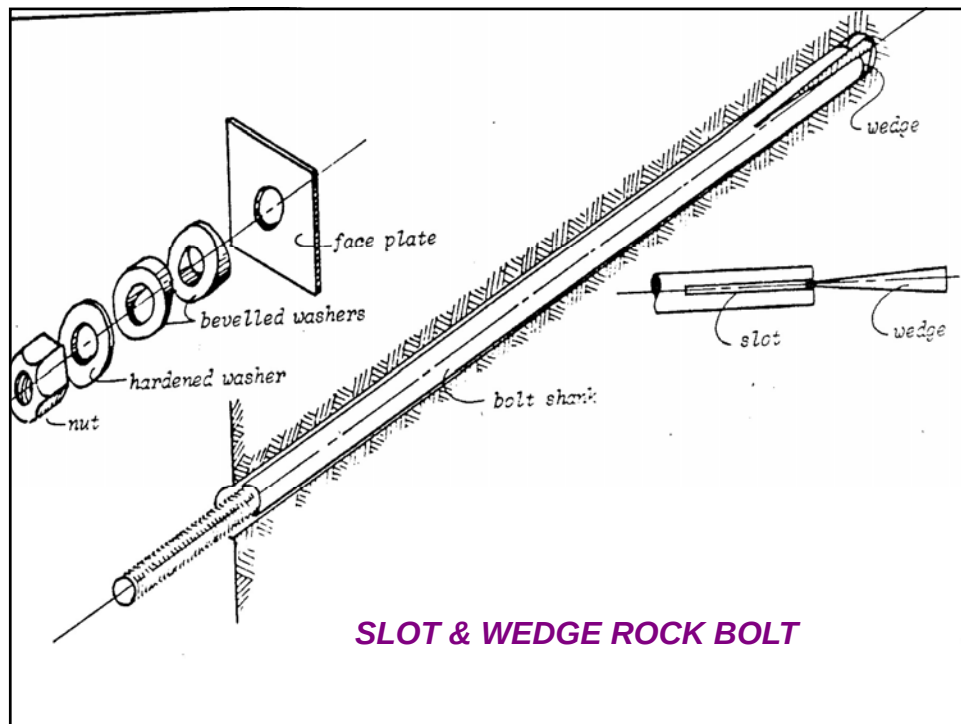
BOLT MATERIAL

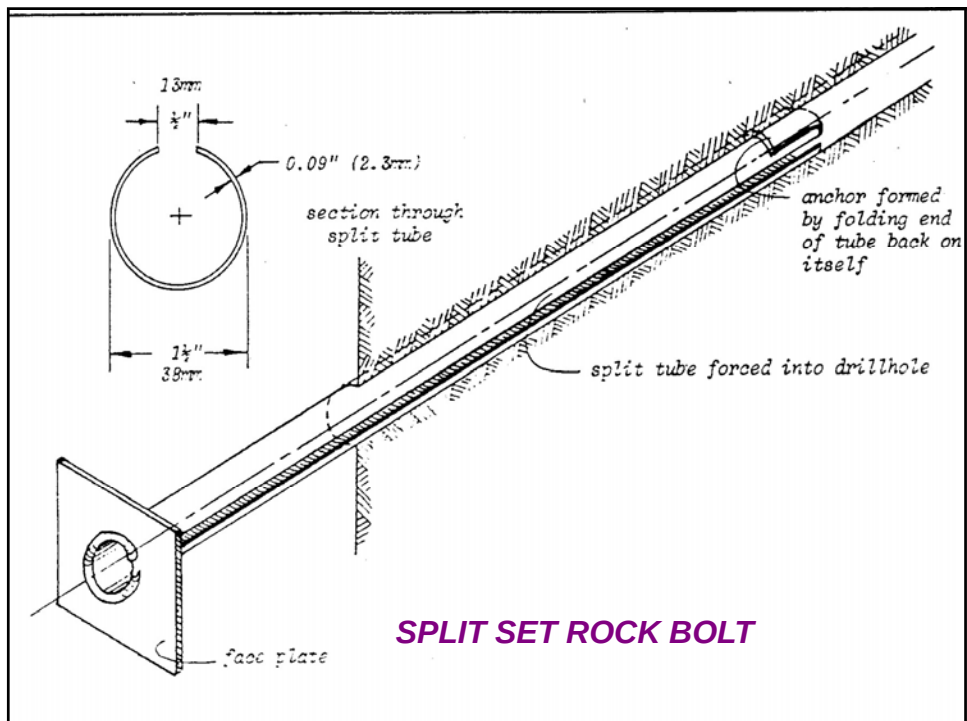
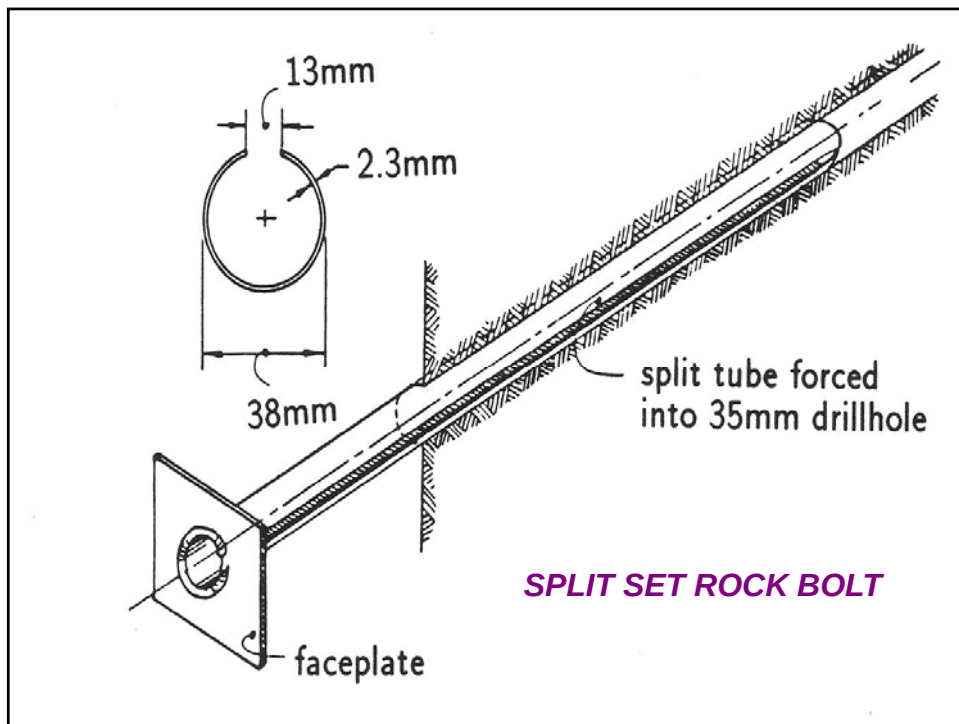


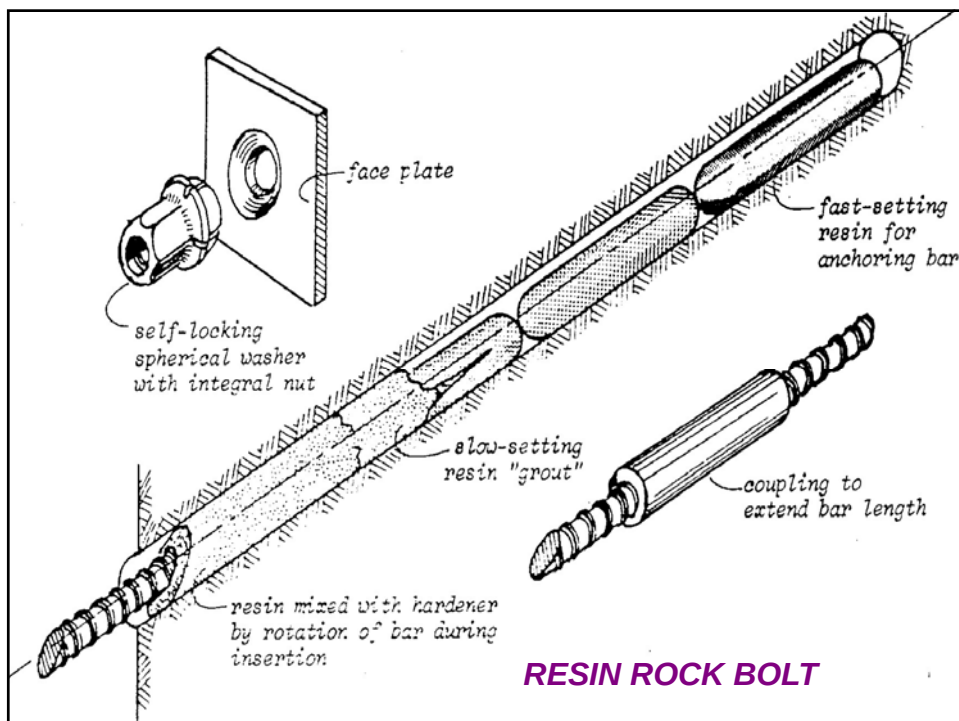
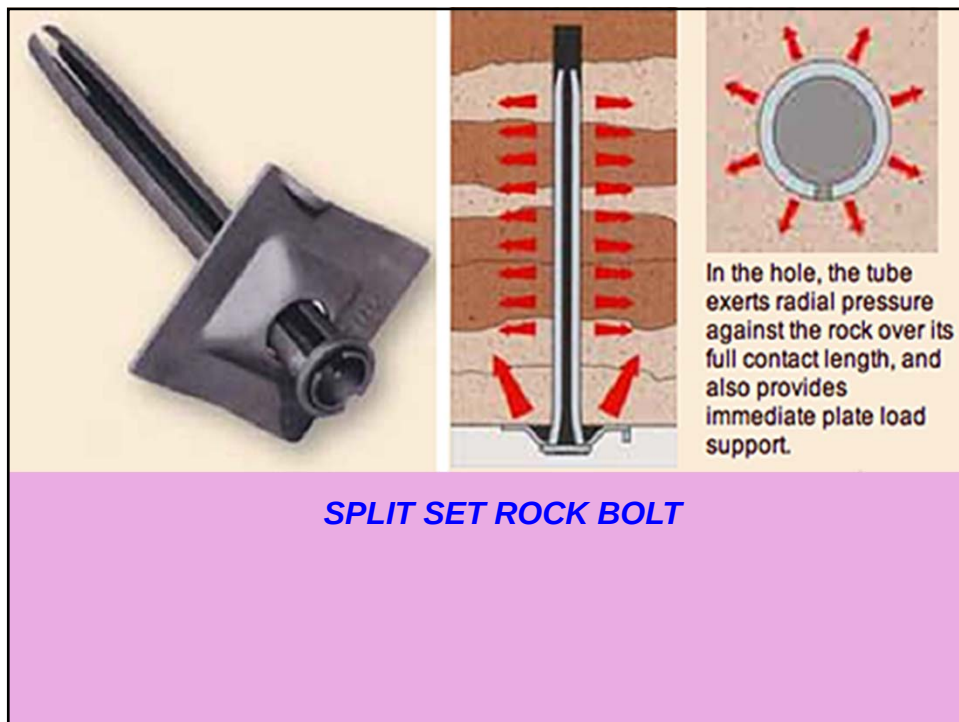
BOLT RIGIDITY

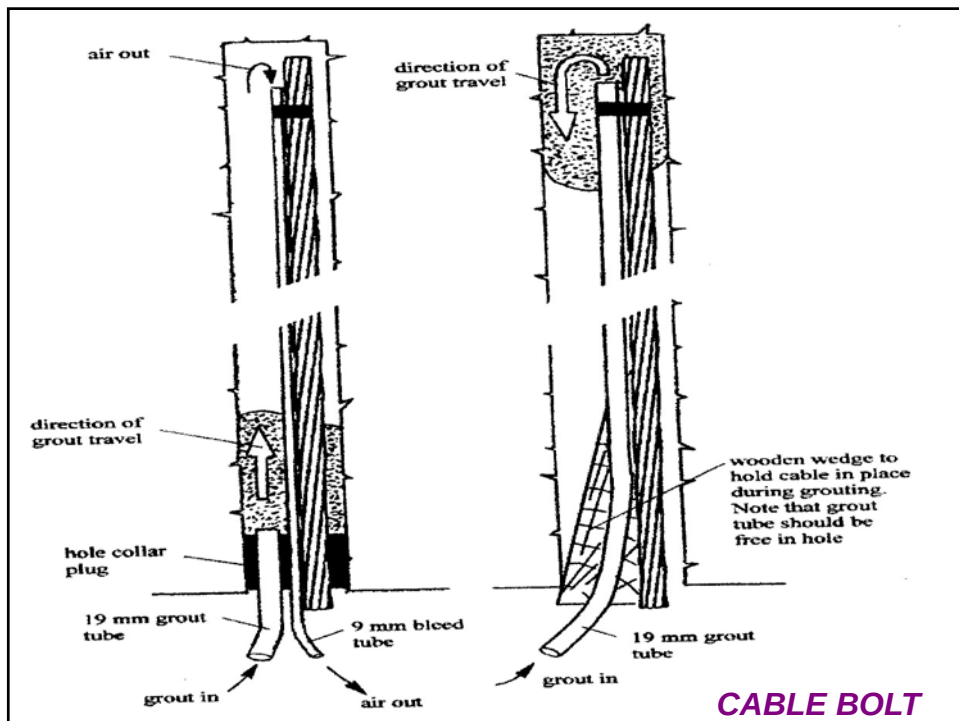
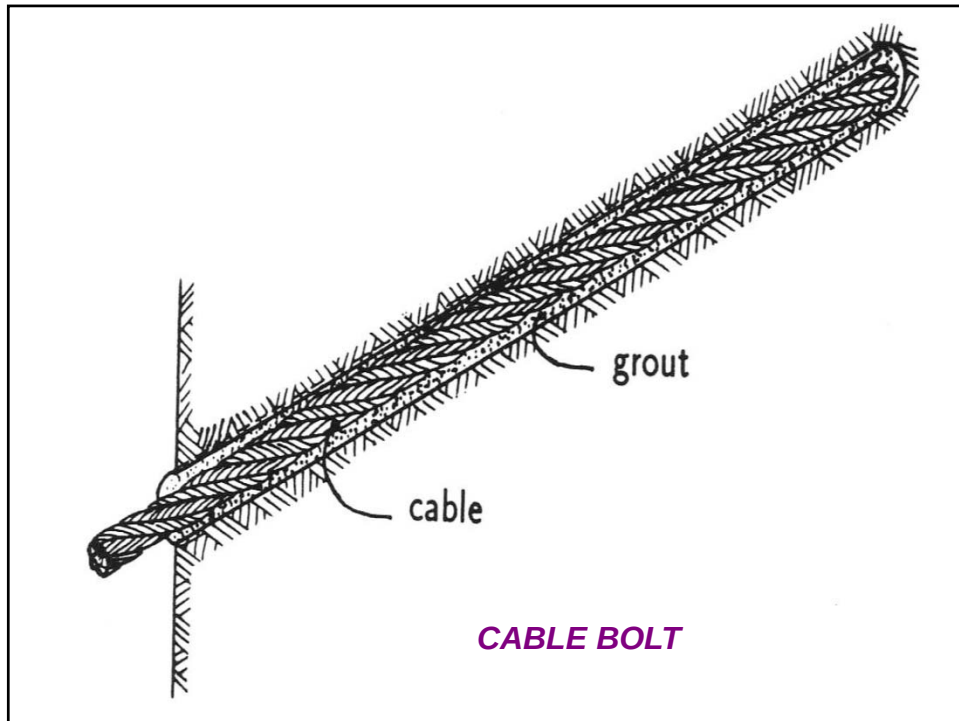


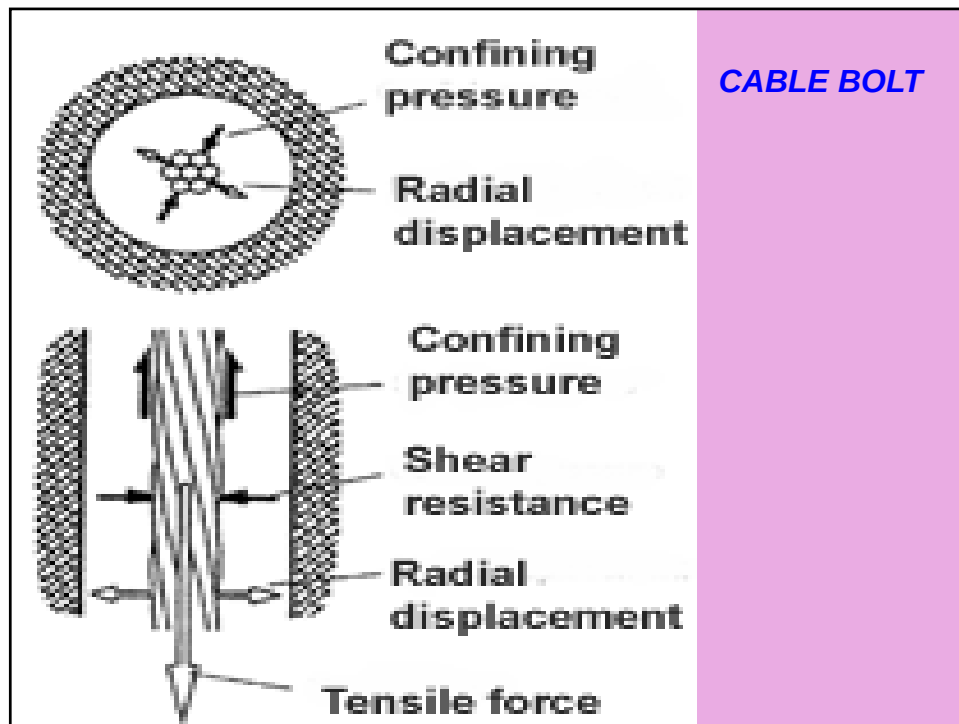


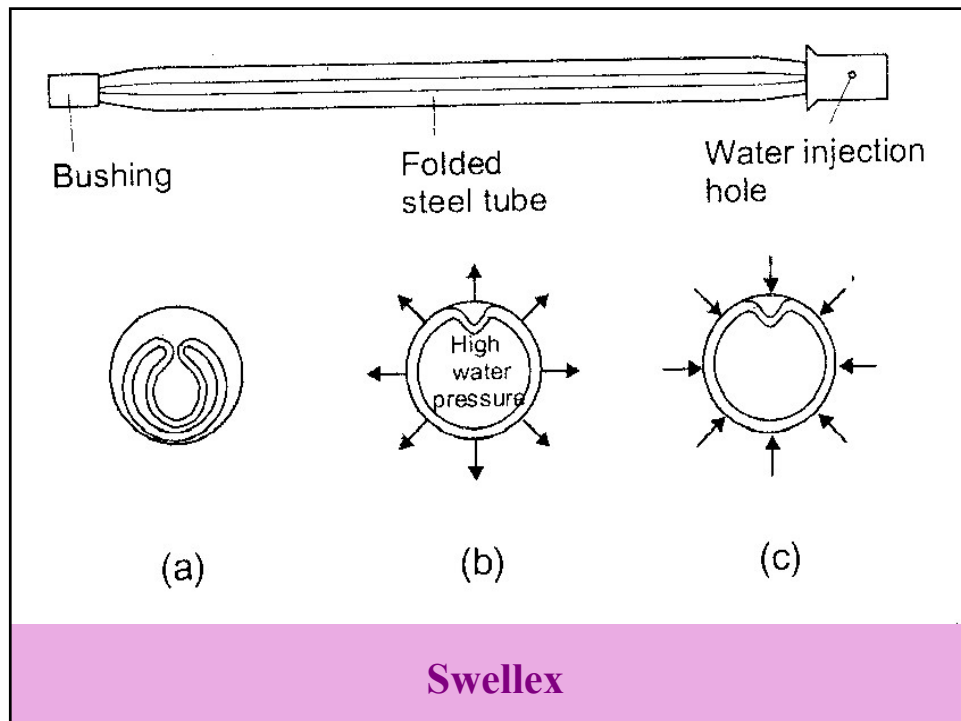
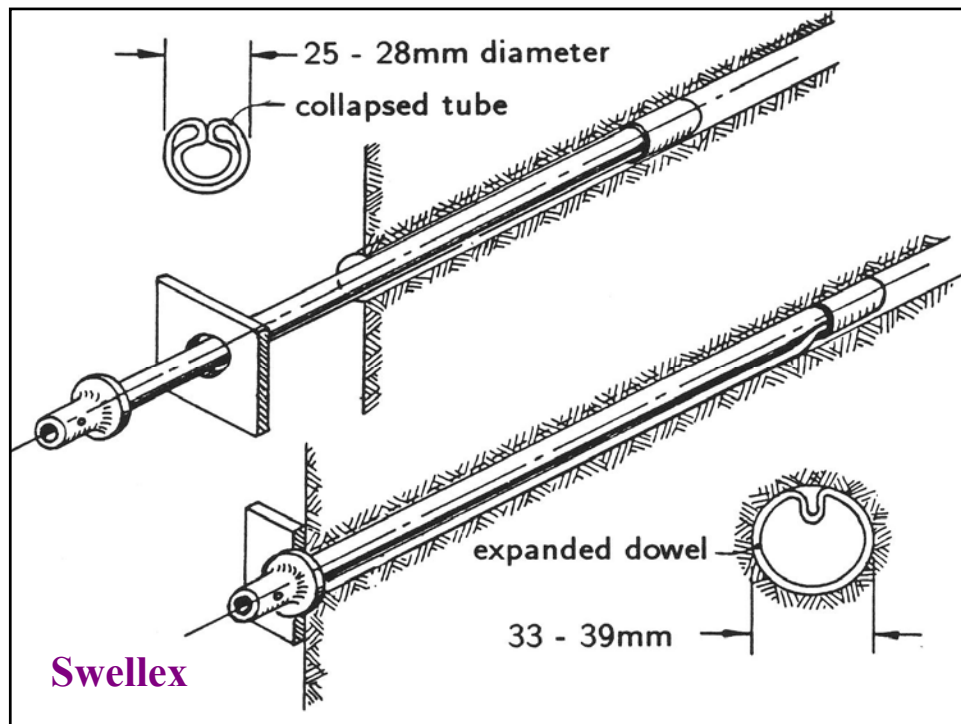


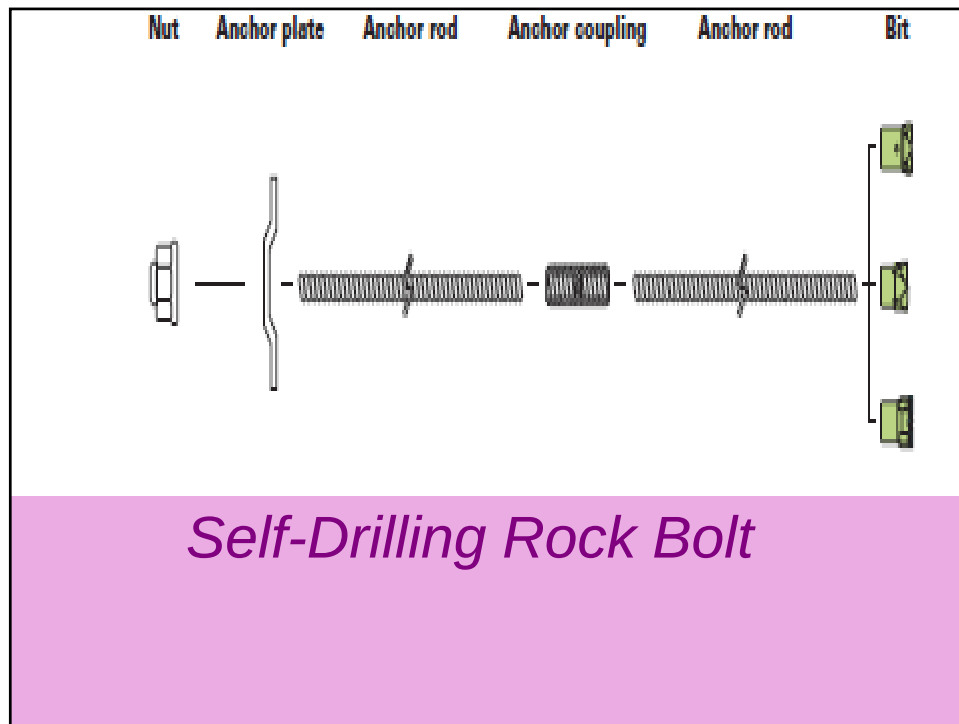


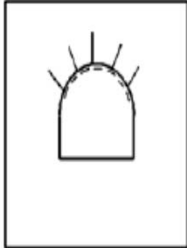
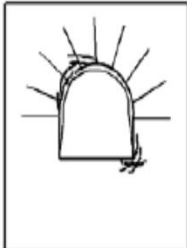


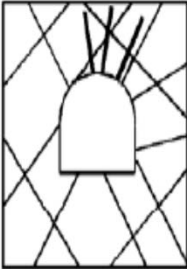








	Low stress levels	High stress levels
Massive rock	 Massive rock subjected to low in situ stress levels. No permanent support. Light support may be required for construction safety.	 Massive rock subjected to high in situ stress levels. Pattern rockbolts or dowels with mesh or shotcrete to inhibit fracturing and to keep broken rock in place.

	Low stress levels	High stress levels
Jointed rock	 Massive rock with relatively few discontinuities subjected to low in situ stress conditions. 'Spot' bolts located to prevent failure of individual blocks and wedges. Bolts must be tensioned.	 Massive rock with relatively few discontinuities subjected to high in situ stress conditions. Heavy bolts or dowels, inclined to cross rock structure, with mesh or steel fibre reinforced shotcrete on roof and sidewalls.

	Low stress levels	High stress levels
Heavily jointed rock	 Heavily jointed rock subjected to low in situ stress conditions. Light pattern bolts with mesh and/or shotcrete will control ravelling of near surface rock pieces.	 Heavily jointed rock subjected to high in situ stress conditions. Heavy rockbolt or dowel pattern with steel fibre reinforced shotcrete. In extreme cases, steel sets with sliding joints may be required. Invert struts or concrete floor slabs may be required to control floor heave.

DESIGN OF ROCK BOLT SYSTEM

The important parameters for designing rock bolt system are:

- Choice of bolt
- Length of bolt
- Diameter of bolt
- Spacing of bolt
- Pattern of rock bolt
- Angle of bolting
- Bolt tension

Types of bolt	Types of anchor	Suitable strata type	Comments
Pointed anchor bolt (tensioned)	Slot & wedge	Hard rock	Simple and cheaper
	Expansion shell	Medium hard rock	Reusable bolt and expensive
	Resin grout	All strata especially for weak rock	Good anchorage and expensive
	Quick setting cement cartridge	All strata	Immediate support
Full-length-grouted bolt (non-tension)	Cement injection	Most strata	Simple and inexpensive
	Perfo	Most strata	Simple and effective Relatively expensive
	Quick setting cement cartridge	Most strata	Immediate support
	Resin grout	All strata especially for weak rock	Good anchorage and expensive
Split set	Full column friction	Most strata	Yieldable support Immediate support
Swellex	Full column friction	Weak rock	Immediate support in poor rock condition
Cable bolt	Cement injection	Weak rock	Mining thick coal seam and wide orebody
Wooden dowel	Resin grout	Medium hard rock	Simple and cuttable
Bamboo dowel	Slot & wedge	Medium hard rock	Simple and cuttable

Concrete Linings

Concrete tunnel linings are divided into **segmental** and **cast in situ** categories which can be further subdivided into **general lining** groups. The majority of concrete linings are used in civil engineering tunnels as either **primary or secondary supports**.

Concrete Linings

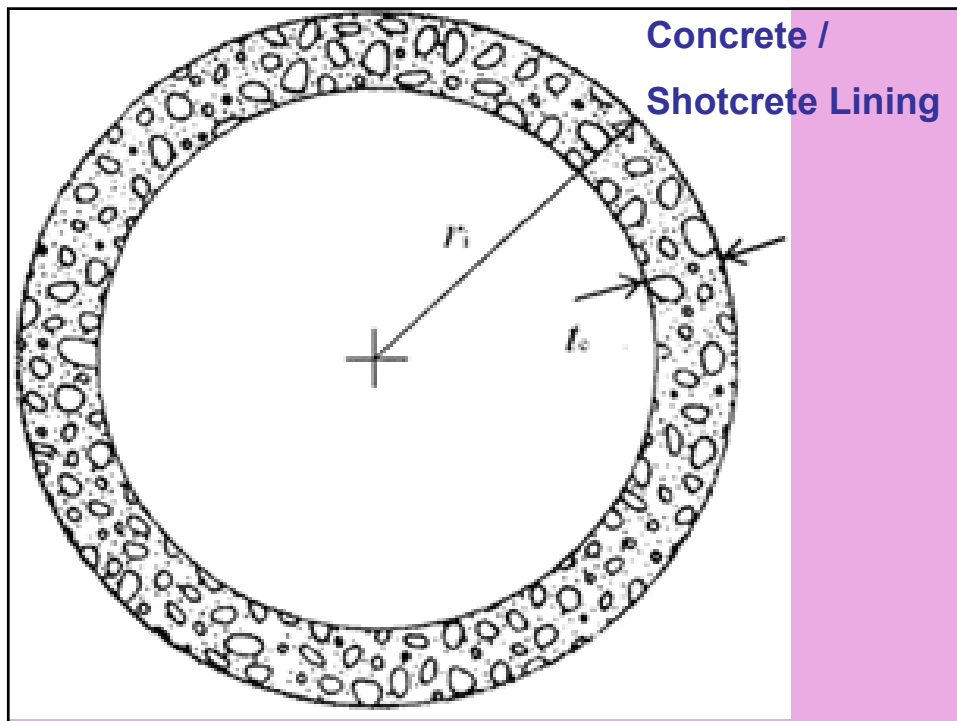
Concrete possesses favourable strength in compression and has a high modulus of elasticity. Its castability and relatively low specific weight and favourable cost when compared to cast iron and steel generally are regarded as important features which promote its popularity for tunnel linings. A major factor influencing its suitability for tunnel linings is that of its durability.

Concrete Linings

Unfavourable features in Concrete

It is weak in tension which results in several considerations which need to be taken into account in respect of tunnel support. This feature is of special importance regarding the use of segmental supports owing to the risk of damage during handling and erection.

The degree of steel reinforcement is required to allow the segments to possess adequate strength to resist cracking or failure due to ground loading in addition to resisting the loads transmitted by the shield shove jacks.



Shotcrete Linings

Shotcrete is a widely used method for tunneling support. This type of rock support is obtained by spraying concrete on the rock surface. Shotcrete for rock support has been used for several decades and has become increasingly popular because of its favorable properties together with high capacity and flexibility.

Shotcrete Linings

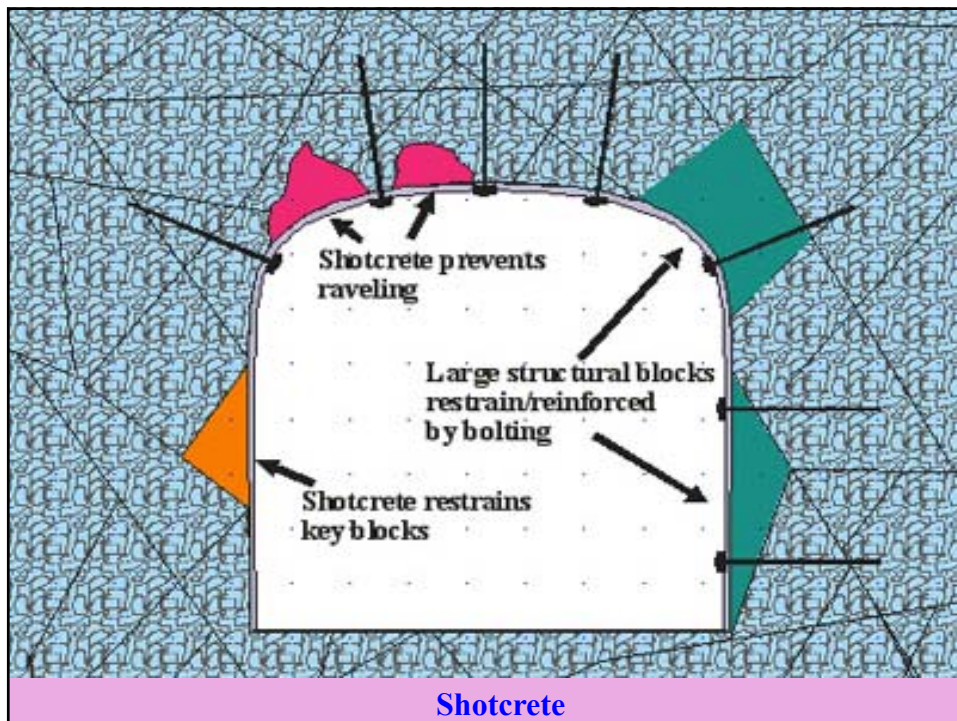
Shotcrete is the generic name for cement, sand and fine aggregate concretes which are applied pneumatically and compacted dynamically under high velocity.

Shotcrete has found widespread use as a temporary support in rock tunnels in conjunction with rock bolting and wire mesh.

Its principal role is that of controlling short term closure and promoting early control of freshly exposed rock faces within the tunnel drive until erection of the primary support in relaxed ground some distance behind the tunnel face.

Shotcrete Linings

Shotcreting or sprayed concrete involves the application of a concrete lining onto a tunnel wall by the action of pumping at a high pressure and spraying through a nozzle which results in impacting onto the rock surface. Such linings exactly match the tunnel profile, and in theory no voids should be left behind the lining. Shotcrete is normally applied to either squeezing or loosening ground.



Shotcrete Linings

In squeezing ground it is used as a temporary support to allow controlled ground movement in the distressing process prior to the installation of a primary support.

In loosening ground conditions, its supporting function is that of providing immediate support to control, if not prevent, loosening that can cause instability in the tunnel heading or result in the generation of excessive loading acting on the tunnel lining.

Shotcrete

TYPICAL SHOTCRETE MIX

<u>Item</u>	<u>% by weight</u>
Cement	15-20
Coarse aggregate	30-40
Fine aggregate/sand	40-50

WATER –CEMENT RATIO

0.3-0.5 in dry mix

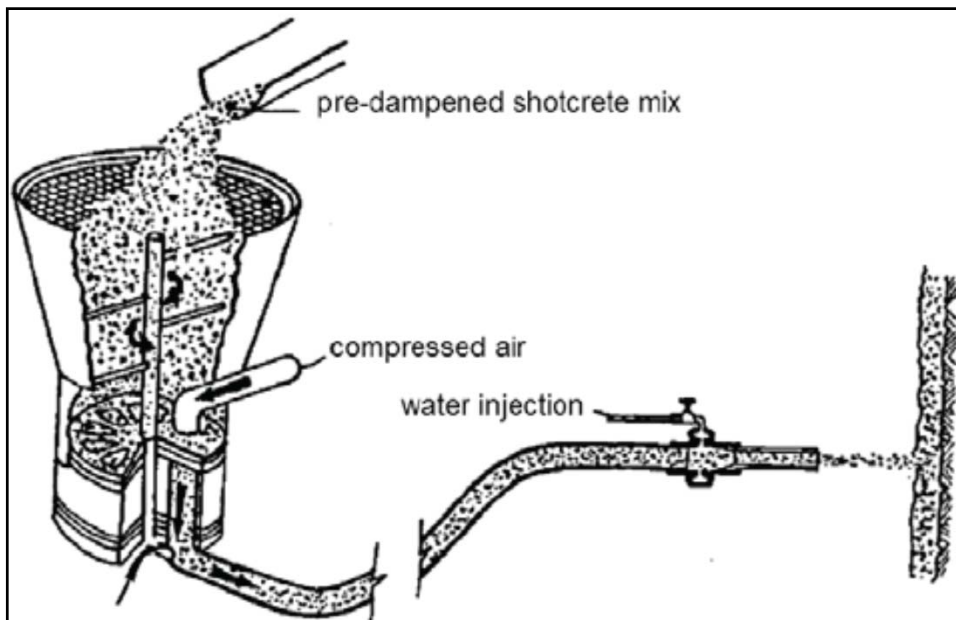
0.4-0.6 in wet mix

Shotcrete can be defined as consisting of material greater than 10 mm in size, whereas the term gunite relates to material less than 5 mm size. Gunite is generally used solely as a tunnel sealant.

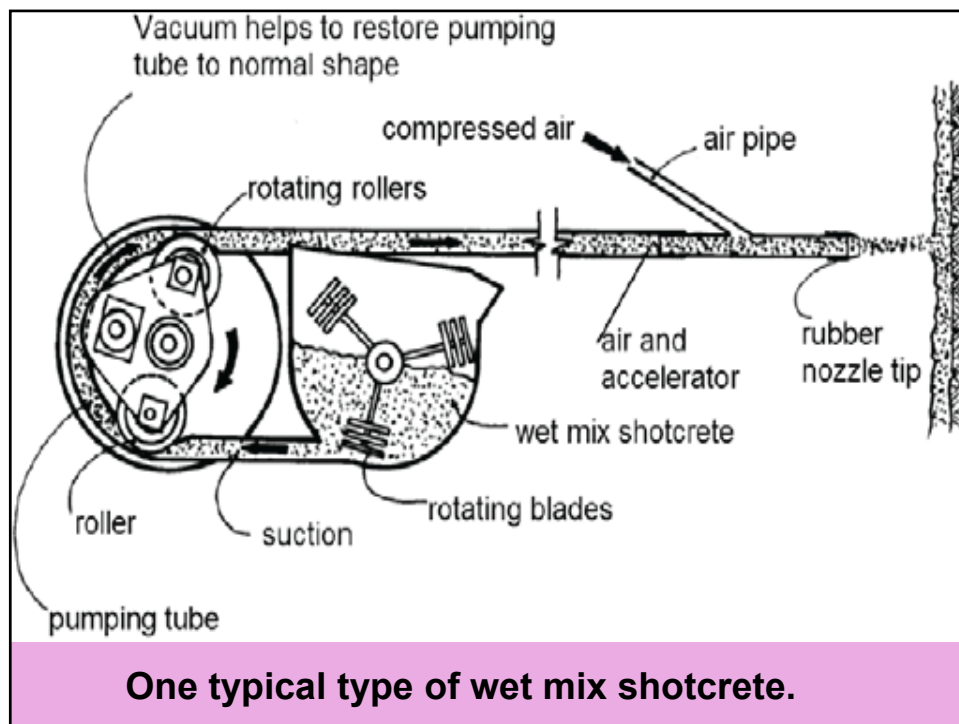
Shotcrete also protects the rock mass from any deterioration effects resulting from contact with the tunnel atmosphere. Typical strength values for shotcrete are around 40-55 MPa, UCS, although higher strengths can be achieved if necessary.

Depending on its requirement the following types of shotcrete is used:

- *Dry Mix Shotcrete*
- *Wet Mix Shotcrete*
- *Fiber-Reinforced Shotcrete*
- *Mesh-Reinforced Shotcrete*



Simplified sketch of a typical dry mix shotcrete system.



Shotcrete

The main advantages of shotcrete as rock support are

- Ready to use on short notice (batching plant on site)
- Framework is not needed
- Applicable in uneven excavation profile
- Easy to combine with other support methods
- Flexible deformation properties.

Shotcrete

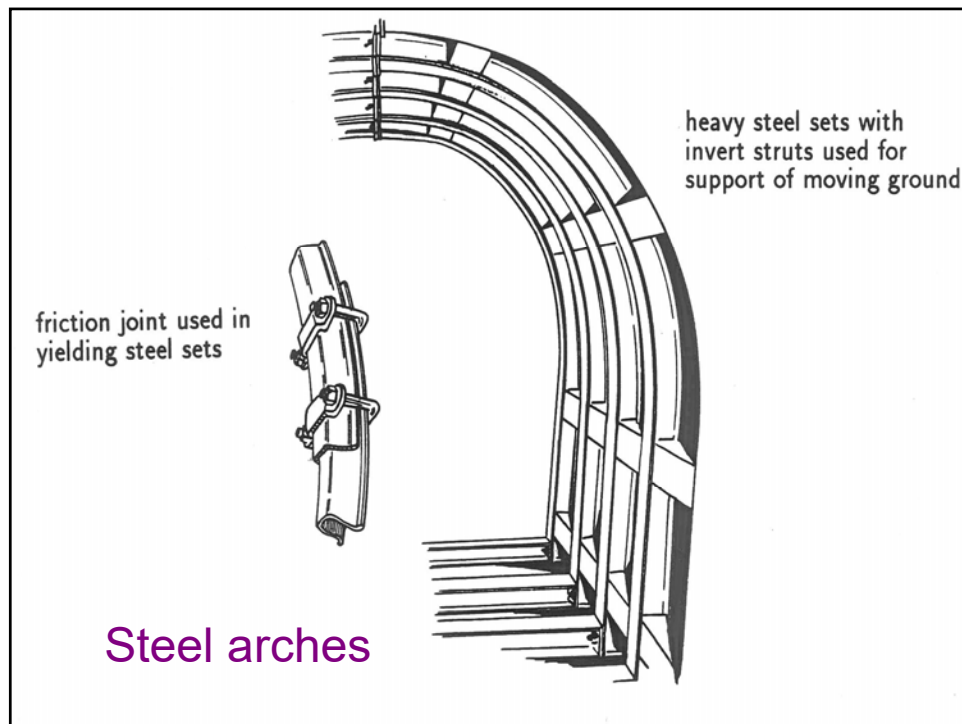
The main disadvantages are:

- Low tension strength (can be improved by mixing fibers or reinforce with casting steel)
- Can collapse when applied on swelling rock types
- Hard to apply on rock types with low cohesion (some types of sedimentary rock)
- Hard to apply on rock with flowing water.

Cast Iron and Steel supports

Metallic linings have proved to be probably the most versatile of all tunnel support systems in view of the wide range of ground conditions and tunnel types in which they have been applied. Cast iron preformed linings were the standard primary support for soft ground tunnels prior to World War II and they still find use in certain situations today.

Similarly, preformed steel supports in the form of rolled steel joist (RSJ) in either H- or U- sections find a wide range of application varying from temporary support systems for rock tunnels in poor quality rock where rock bolting is considered inapplicable.



Steel Linings

Steel linings offer the advantages of being equally strong in both compression and tension and thus, are able to withstand high bending moments whilst possessing favourable post yield characteristics. In respect of the latter property, such linings retain appreciable resistance to movement beyond the yield condition particularly under conditions of significant radial deformation. Consequently, they are suited to ground conditions which are likely to induce significant closure due to the support system being subjected to stress induced progressive rock failure

Cast Iron Lining

Cast iron preformed linings tend to be limited to circular tunnels as their use in soft ground is normally in association with a shielded tunnelling method whilst a closed-loop lining is required for waterproofing purposes. The segmental cast iron lining occurs in various forms and has essentially been a major choice for support in soft ground tunnelling prior to WWII. The main reasons for its choice have been its ease of castability into segments, favourable corrosion resistance and compressive strength, relatively high modulus of elasticity and attractive support cost when compared to brickwork and masonry linings which it replace in the late 19th Centaury.

Cast Iron Lining

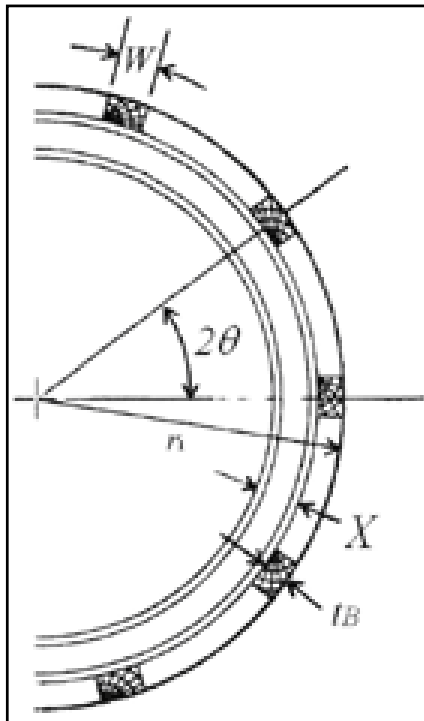
Since the 1940's concrete segmental linings have largely superseded it. However, cast iron linings are employed at present day specifically in medium to large diameter tunnels in area where waterproofing is a major factor. Cast iron linings are still recognised as being more adaptable to waterproofing than concrete segmental linings.

Rolled Steel Joist Support (RSJ)

Steel supports are adaptable to almost any shape of opening in either open or closed loop due to their strength in both compression and tension and the ability of rolling steel linings to virtually any given shape. The most commonly used tunnel profiles with steel supports are circular, arch/horseshoe shaped and rectangular/square.

Rolled Steel Joist Support (RSJ)

The most common form of RSJ used in tunnelling as temporary support is that of the rigid H or I – section. It is usually subsequently encased in concrete cast in situ after the short term deformations associated with the excavation process have occurred and correspondingly accepted by the steel supports.



Blocked Steel Set

Rolled Steel Joist Support (RSJ)

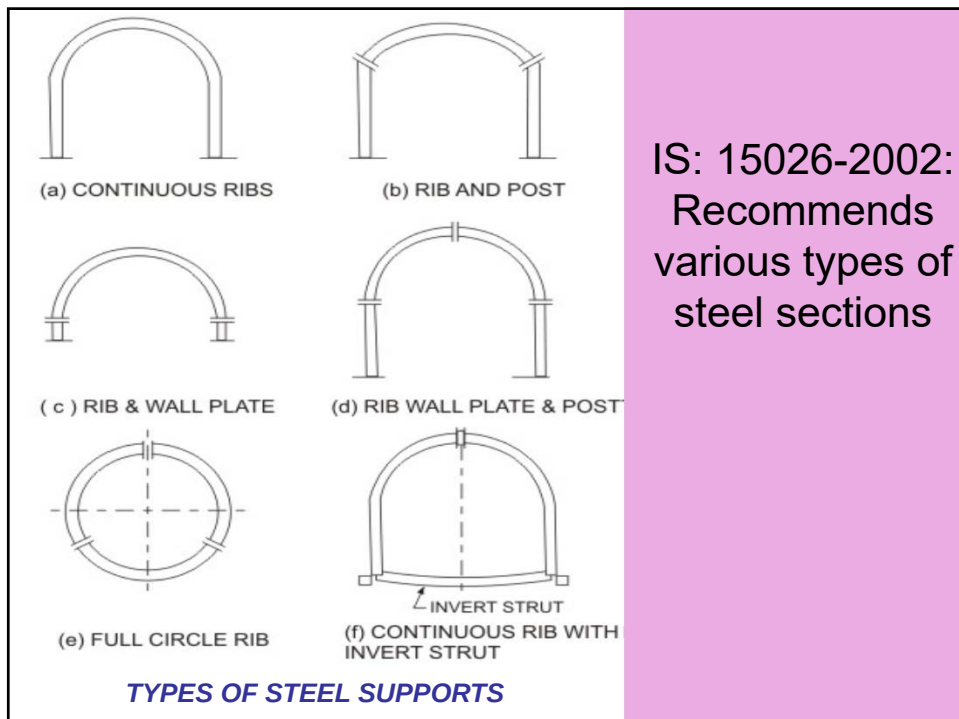
In civil engineering tunnels where an aesthetic finish is required, temporary steel supports are often used in conjunction with cast in situ concrete with the steel supports being employed to accommodate the initial ground deformations thus facilitating controlled stress relaxation and minimising the deformations imposed on the rigid concrete lining. Such action is essential due to the rigid concrete lining's poor ability to sustain high bending moments without fracture, and the final profile of such tunnels usually masks the true deformation history of the steel supports under the concrete lining.

Rolled Steel Joist Support (RSJ)

IS: 15026-2002 recommends various types of steel ribs, as follows:

- a) Continuous rib
- b) Rib and post
- c) Rib and wall plate
- d) Rib, wall plate and post
- e) Full circle rib
- f) Invert strut with continuous rib

These are built of steel sections either shaped or welded in pieces in the form of a curve or a straight section.



Grouting

This is a cement mortar with proportion of cement, sand and water in the ratio 1:1:1 by weight usually, though it may be modified suitably according to site conditions.

Grouting is carried out to fill discontinuities in the rock by a suitable material so as to improve the stability of the tunnel roof or to reduce its permeability or to improve the properties of the rock.

Grouting

Grouting is very rarely used in the tunnels or caverns except in cases of high water inflow or tunnelling is very difficult or there is a risk of washing out of in-filled materials. Pre-grouting can be used in major faults and weakness zones to improve the strength of rock and seepage of water in the rock. Pre-grouting (ahead of the face) normally gives better results than post-grouting (behind the face). The grouting is carried out around the concrete plug in pressure tunnel and pressure shaft extensively.

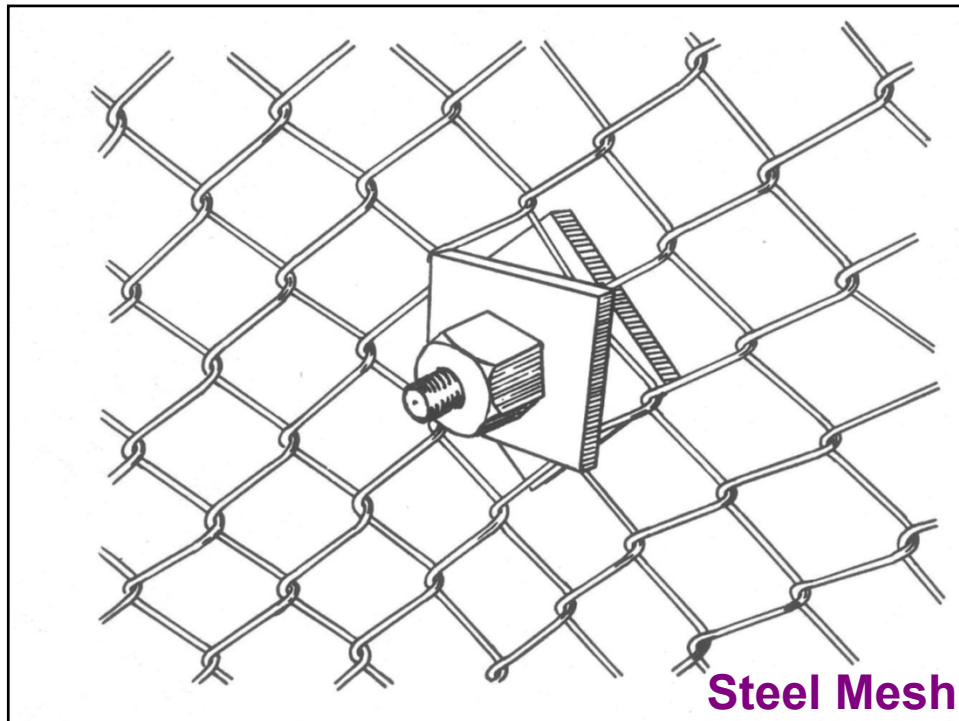
Grouting

The basic design requirement generally involve the following:

- a) Filling the voids, cavities, between the concrete lining and rock and /or between the concrete and steel liner;
- b) Strengthening the rocks around the opening by filling up the joints in the rock system;
- c) Strengthening the rock shattered around the opening;
- d) Strengthening the rock, prior to excavation by filling the joints with cementing material and thus improving its stability; and
- e) Closing water bearing passages to prevent the flow of water into the tunnel and/or to concentrate the area of seepage into a channel from where it can be easily drained out.

Steel Mesh

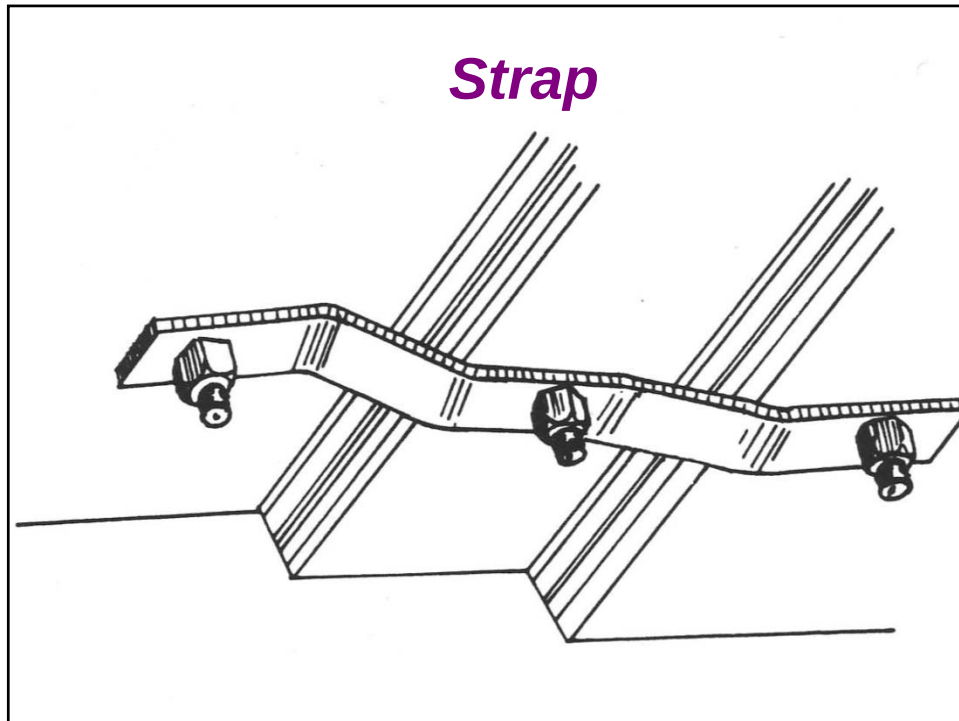
Steel mesh is a form of lagging used behind the main steel structural members, usually steel arches, for the purpose of controlling the detachment and movement of smaller pieces or rock formed at the periphery of excavation.



Strap

This system is used extensively in civil engineering applications to hold slabby ground between rock bolts or to prevent slabs from loosening.

- Straps are easy to install with rock bolts. They are very effective in stratified, slabby ground.
- Straps cannot be used to control ravelling in loose, blocky ground if the block size is small.
- Steel straps are used to give roof control in ground where rockbolts or dowels alone cannot hold intermediate and relatively thin slabs of rock. The strap is formed to follow the rock surface and held with regular rock reinforcement or specially installed pins. Straps should always be installed across weakness planes.

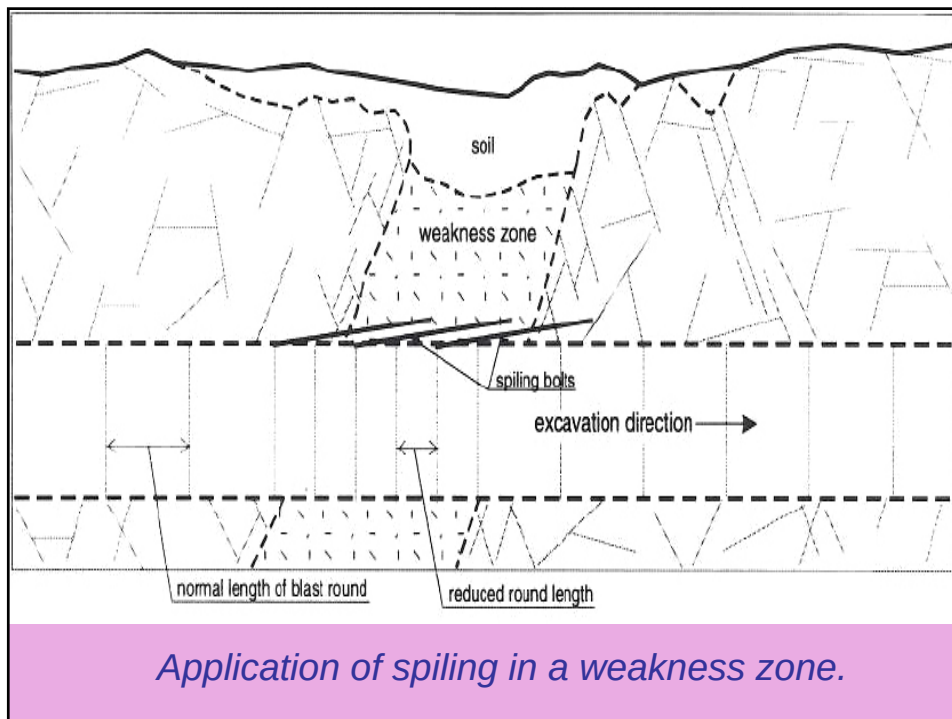


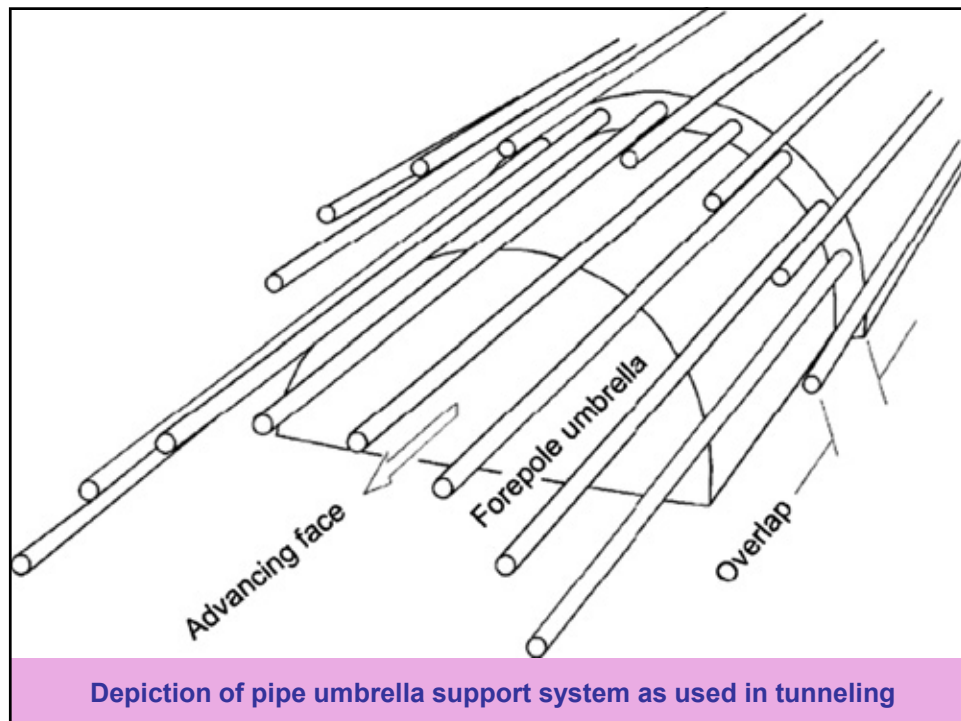
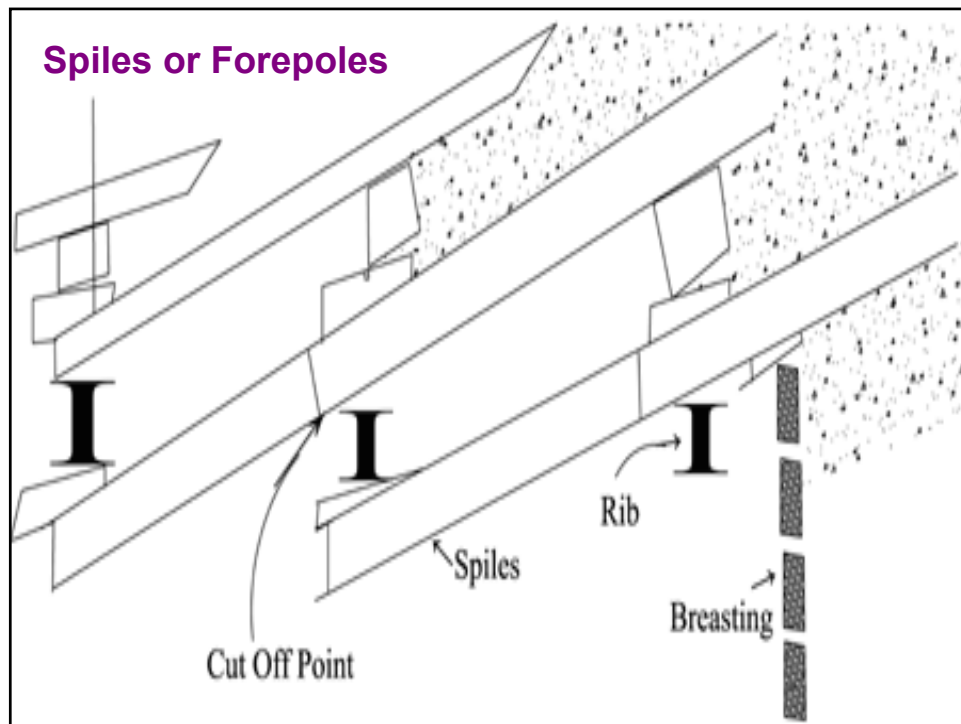
Other forms of Steel Support Lagging

- There various forms of lagging available for use around steel supports. These include wooden boards, corrugated steel sheets, and concrete slabs.
- The choice of logging depends on the life of the tunnel, the nature of the rocks surrounding tunnel, presence of type of geological discontinuities, and the degree of performance required of the support system.

Spiles or Forepoles

Spiles or forepoles are used interchangeably to describe support elements consisting of pipes or pointed boards or rods driven ahead of the steel sets or lattice girders. These elements (herein called spiles) provide temporary overhead protection while excavation for and installation of the next set or girder is accomplished. Typically, spiles are driven in an overlapping arrangement so that there is never a gap in coverage.





Support Section Criteria

The choice of support is made based on the specific requirement of a particular situation such as

- i) Convergence of the opening
- ii) Load on support (unit load, load density)
- iii) The type of rock support that has to be provided for a tunnel depends upon the type of rock quality, which is classified according to its behaviour when an opening is made in the rock such as:
 - LOAD-YIELD CHARACTERISTIC OF ROCKMASS SURROUNDING THE OPENING
 - PHYSICO-MECHANICAL PROPERTIES OF ROCK MASS
 - TIME DEPENDENT BEHAVIOUR OF ROCK MASS
 - DISCONTINUITIES IN ROCK MASS
 - ANISOTROPY IN ROCK MASS
 - ENVIRONMENT IN EXCAVATION LOCATION

Support Section Criteria...

- vi) Degree of mechanization
- vii) Rate of face advance
- viii) Method of excavation
- ix) Type of tunnel instability
- x) Functional requirement of the tunnel
- xi) Life of tunnel and support
- xii) Factor of safety

Support Section Criteria...

xiii) The type of support that has to be provided for a tunnel depends upon the type of material and structural properties of support such as:

- Load-yield characteristic of support
- PHYSICO-MECHANICAL PROPERTIES OF SUPPORT
- TIME DEPENDENT BEHAVIOUR OF SUPPORT
- BEHAVIOUR OF SUPPORT IN UNDERGROUND ENVIRONMENT

xiv) Support installation time and rate

xv) Shape/profile of excavation

xvi) Magnitude and direction of insitu stress

xvii) Stresses at the boundary of excavation

xviii) Support cost

- CAPITAL COST
- INSTALLATION COST
- CONSUMABLE COST
- COST OF MAINTENANCE

DESIGN OF SUPPORT AND REINFORCEMENT SYSTEM

One of the main goals of a designer is to determine the most suitable and economical support system with a convenient excavation method.

The design process in support & reinforcement is often complicated due to the lack of control over the geological and stress environments.

DESIGN OF SUPPORT AND REINFORCEMENT SYSTEM

There are no universal standard analyses for determining rock support design, because each design is specific to the circumstances (scale, depth, presence of water, etc.) at the actual site and the national regulations and experience. Support design for a tunnel in rock often involves problems that are of relatively little or no concern in most other branches of solid mechanics.

DESIGN OF SUPPORT AND REINFORCEMENT SYSTEM

The material and the underground opening forms an extremely complex structure. *"It is seldom possible, neither to acquire the accurate mechanical data of the ground and forces acting, nor to theoretically determine the exact interaction of these"*

Therefore, the rock engineer is generally faced with the need to arrive at a number of design decisions in which judgement and practical experience must play an important part.

Design Criteria

Design criteria are specified by four parameters:

- A load density (tons per foot of entry) required to support the roof,
- a maximum allowable convergence or
- roof displacement (to preserve roof integrity),
- the entry width and the entry height.

The performance of a given support is determined by the support design (size or model) and the support layout (spacing and number of rows across the entry).

Design Criteria

For a given support design, the layout is adjusted to meet the specified design criteria (primarily spacing, for standing supports and number of rows, for intrinsic supports).

The support design is optimized by finding the minimum design (smallest dimensions or smallest model) that meets the load density criteria and maximum convergence (displacement) criteria.

SUPPORT AND REINFORCEMENT DESIGN METHODS

- **EMPIRICAL METHODS**
- **ROCK- SUPPORT INTERACTION METHOD
(GROUND REACTION CURVE)**
- **OBSERVATIONAL METHODS**
- **EMPIRICAL RULES**
- **PHYSICAL MODELLING METHODS**
- **ANALYTICAL AND NUMERICAL METHODS**

SUPPORT AND REINFORCEMENT DESIGN METHODS

The choice of suitable tools for the design is essentially an outcome of the actual ground behaviour, such as an acceptable standard or some other requirement.

EMPIRICAL METHODS OF DESIGN

- Rock mass classifications form the backbone of the empirical design approach and are widely employed in rock engineering
- In fact, on many projects, the classification approach serves as the only practical basis for the design of complex underground structures

Major rock classifications currently in use

Name of classification	Originator and date	Country of origin	Applications
Rock loads	Terzaghi, 1946	USA	Tunnels with steel supports
Rock Structure Rating (RSR)	Wickham et al. 1972	USA	Tunnelling
Geomechanical classification (RMR) system	Bieniawski, 1973	S. Africa & USA	Tunnels, mines, foundations
Q-system	Barton, et al. 1974	Norway	Tunnelling, large chambers
Rock Mass Index (RMi)	Palmstrom. 1995 ,2000	Norway	Tunnelling, Underground Excavations
Geological Strength Index	Hoek and Brown,1997,2000	USA	Tunnelling, Underground Excavations

EMPIRICAL METHODS OF DESIGN

During the Feasibility and preliminary design stage of the project, very little information of rock mass and stress is available. The rock mass classification system can be quite useful at this stage. The classification is used to obtain a general rating of rock mass quality and classification of special rock engineering properties such as drillability and blastability and to estimate the rock supports.

Two commonly used methods are:

1. Q- method
2. RMR method

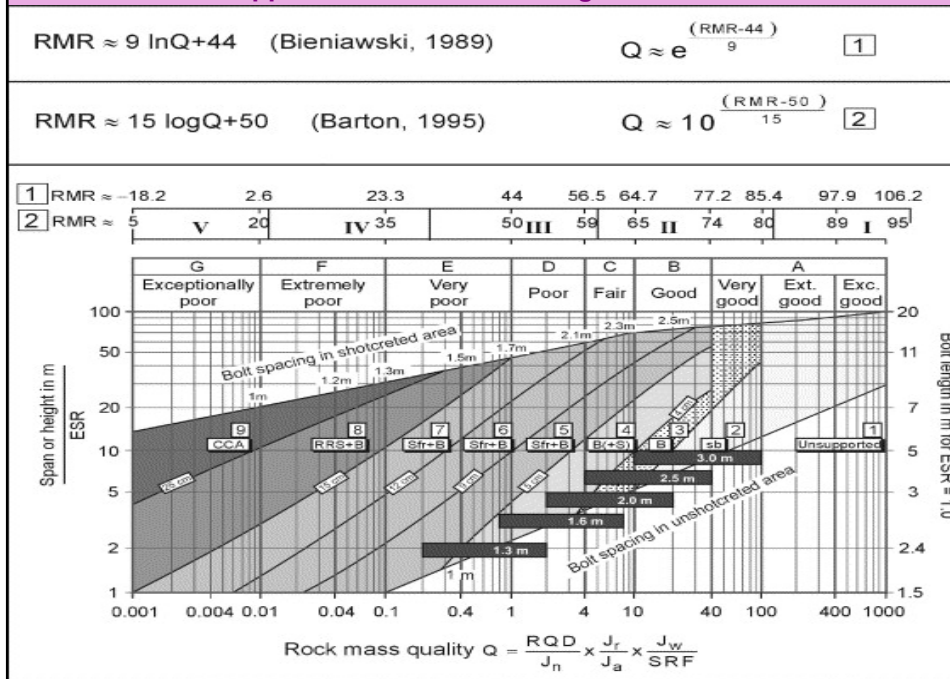
PARAMETERS IN ENGINEERING OF SOME COMMONLY USED ROCK MASS CLASSIFICATIONS

Engg. Rock Mass Classification	Major Parameters	Rock Rating and Load on Supports (P)
Geomechanical classification (RMR) system	1) Intact rock strength, 2) Rock Quality Designation (RQD) 3) Spacing of discontinuities 4) Condition of discontinuities: 5) Groundwater 6) Discontinuity orientation (adjustment)	RMR = Ratings of Parameters (1 + 2 + 3 + 4 + 5 - 6) $P = \frac{100 - RMR}{100} \times \gamma \times B$ $\gamma = \text{Density of Rock}$ $B = \text{Width of Excavation}$
Q-system	1) Rock Quality Designation (RQD) 2) Joint Set Number. (J_n) 3) Joint Roughness Number (J_r) 4) Joint Alteration Number (J_a) 5) Joint Water Reduction Number (J_w) 6) Stress reduction Factor (SRF)	$Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$ $P = \frac{2}{J_r} \left(Q^{-\frac{1}{3}} \right)$

Table: Guidelines for excavation and support of 10 m span rock tunnels in accordance with the RMR system (After Bieniawski 1989)

Rock mass class	Excavation	Rock bolts (20 mm diameter, fully grouted)	Shotcrete	Steel sets
I - Very good rock RMR: 81-100	Full face, 3 m advance.	Generally no support required except spot bolting.		
II - Good rock RMR: 61-80	Full face, 1-1.5 m advance. Complete support 20 m from face.	Locally, bolts in crown 3 m long, spaced 2.5 m with occasional wire mesh.	50 mm in crown where required.	None.
III - Fair rock RMR: 41-60	Top heading and bench 1.5-3 m advance in top heading. Commence support after each blast. Complete support 10 m from face.	Systematic bolts 4 m long, spaced 1.5 - 2 m in crown and walls with wire mesh in crown.	50-100 mm in crown and 30 mm in sides.	None.
IV - Poor rock RMR: 21-40	Top heading and bench 1.0-1.5 m advance in top heading. Install support concurrently with excavation, 10 m from face.	Systematic bolts 4-5 m long, spaced 1-1.5 m in crown and walls with wire mesh.	100-150 mm in crown and 100 mm in sides.	Light to medium ribs spaced 1.5 m where required.
V - Very poor rock RMR: < 20	Multiple drifts 0.5-1.5 m advance in top heading. Install support concurrently with excavation. Shotcrete as soon as possible after blasting.	Systematic bolts 5-6 m long, spaced 1-1.5 m in crown and walls with wire mesh. Bolt invert.	150-200 mm in crown, 150 mm in sides, and 50 mm on face.	Medium to heavy ribs spaced 0.75 m with steel lagging and forepoling if required. Close invert.

Table: Support selection chart using Q Index and RMR



ROCK- SUPPORT INTERACTION METHOD(GROUND REACTION CURVE)

When an opening is excavated for an underground structure, the existing stresses prior to excavation redistribute and adjust themselves to a new equilibrium condition. These stress changes require displacements to occur and the excavated ground tries to converge toward the opening. The amount of convergence depends on the host ground characteristics, method of excavation and the size of opening used

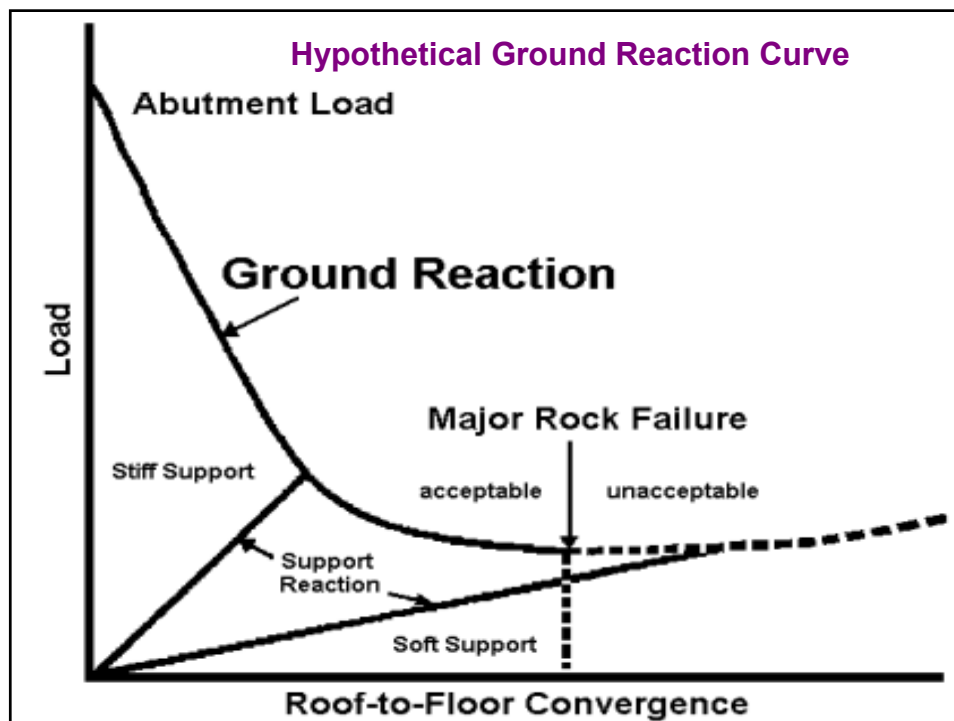
ROCK- SUPPORT INTERACTION METHOD(GROUND REACTION CURVE)

The Ground Reaction Curve is simply a measure of how the ground responds (deforms) relative to the amount of roof support that is provided.

The basic premise is that the magnitude of the support resistance controls the amount of roof-to-floor convergence that will occur.

ROCK- SUPPORT INTERACTION METHOD(GROUND REACTION CURVE)

In general, higher support capacities will be required to reduce the convergence. The reason for this is that the ground sheds load to surrounding mine structures as it deforms, thereby requiring progressively less support to maintain equilibrium. The most important point is that only so much deformation can occur before the roof becomes unstable, leading to a roof fall. Hence, it is critical that sufficient support be installed before this critical deformation is reached.



ROCK- SUPPORT INTERACTION METHOD(GROUND REACTION CURVE)

The goal is to determine a profile of how much convergence occurs when the support resistance changes. Ideally, wants to get a full picture of the ground reaction curve which includes

- ✓ The maximum amount of support necessary to minimize convergence
- ✓ The minimal acceptable support which is the least amount of support necessary to prevent a roof fall from occurring
- ✓ The convergence at the minimal acceptable support approaches that where roof instability is about to occur

ROCK- SUPPORT INTERACTION METHOD(GROUND REACTION CURVE)

As part of this process, an effort should also be made to determine the critical roof-to-floor convergence where roof failure occurs, since this will be a critical design value for the support design. However, in order to do this, a low density support system must be installed that will allow considerable convergence to occur. Since this poses a risk of inadequate ground control, precautions should be taken.

ROCK- SUPPORT INTERACTION METHOD(GROUND REACTION CURVE)

The load-displacement data is used to generate a plot of support load density as a function of convergence (ground reaction curve). Each support type with different stiffness represents one data point on the ground reaction curve. The support load density is determined as the measured load in the support at the observed convergence times the number of supports per unit area of tunnel .

OBSERVATIONAL DESIGN METHODS

Observational methods utilize monitoring as an integral constituent of the design process. Monitoring techniques are also employed to complement empirical, analytical and numerical methods. The aim is to determine ground response to reinforcement, allowing the early identification of possible problem areas and the verification of the implemented design

OBSERVATIONAL DESIGN METHODS

NATM METHOD

- NEW AUSTRIAN TUNNELLING TECHNIQUE (NATM) is a method by which the host ground surrounding an excavation for an underground structure is made into an integral part of the support structure.
- The host ground and the external support structure together take the full load. The host ground takes a major share of the load and the support takes a much smaller share of the ground load. This results in saving costs of external support systems

NATM METHOD

One will notice that the tangential stresses are always higher than radial stresses when an opening is created. Thus, if a support system can provide tangential resistance in the form of increased frictional resistance at the support and host interface, then the further relaxation of stresses due to excavation can be adequately resisted

NATM METHOD

Shotcrete provides strong frictional resistance. The ideal resistance will be provided by a closed ring of a very thin shotcrete membrane. But many times it is not practical to close the invert of the opening by shotcreting. Thus, the shotcrete in the roof and the sidewalls have to provide the tangential resistance. In order to help the resisting capability of this open shotcrete ring thus formed, use of rockbolts become necessary.

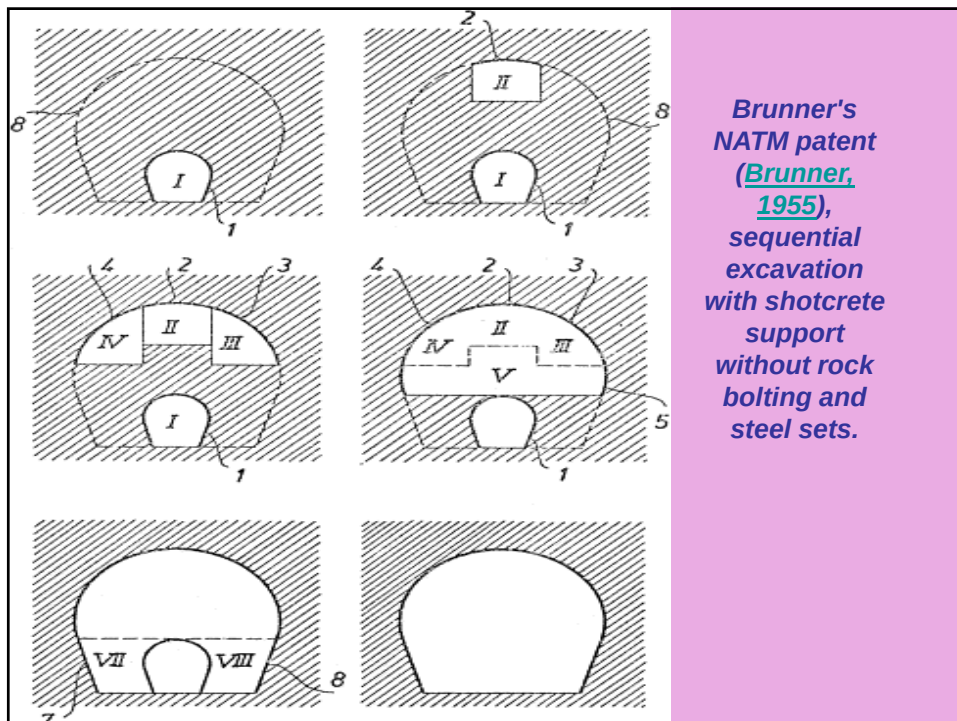
NATM METHOD

The NATM is an observational method and requires

- Application of a thin layer of shotcrete with or without rock bolts, wire mesh fabric, and lattice girder
- Monitoring and observing the convergence of the opening.

NATM METHOD

- If the observed convergence exceeds the acceptable limits, then subsequent applications of next layers of shotcrete are required until the convergence has stopped or is within the acceptable range. The shotcrete thickness is, thereby, optimised according to the admissible deformations.
- The geometry of the opening is very crucial so that adequate ground arching action can develop. Straight reaches are carefully substituted by curved configurations

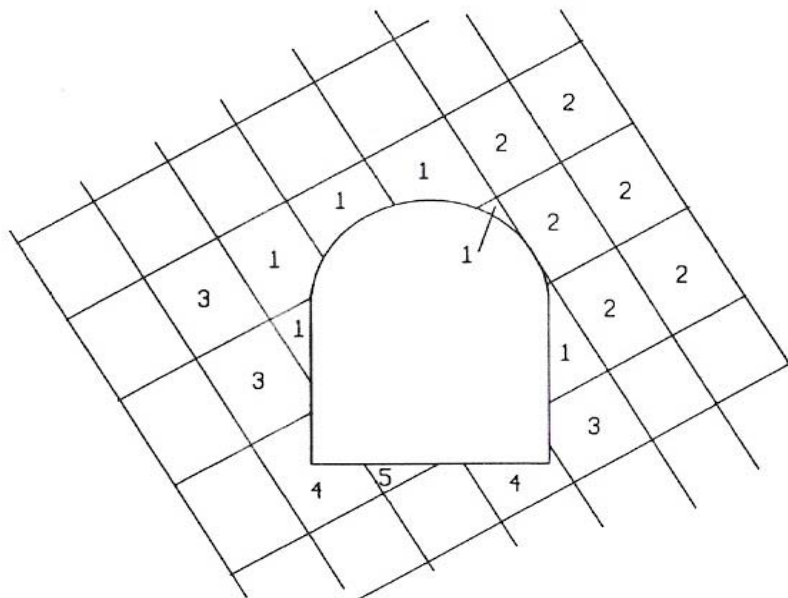


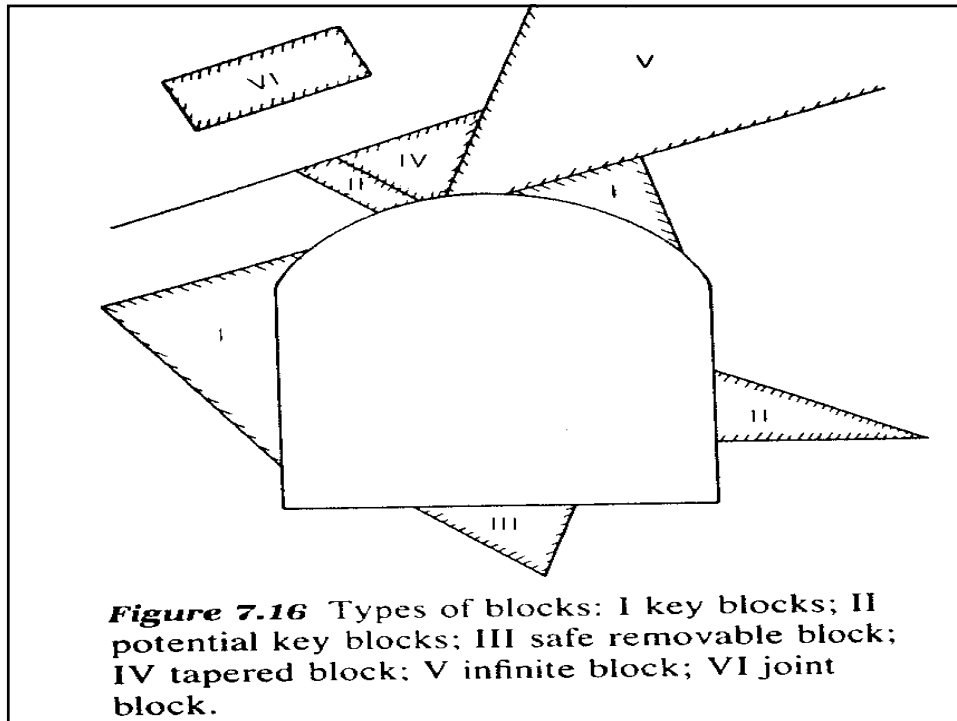
ANALYTICAL METHODS

DISCONTINUITY ANALYSIS METHOD

IT IS WELL KNOWN THAT ROCK CONTAINS DISCONTINUITIES IN THE FORMS OF BEDDING PLANES, JOINTS, FAULTS, FOLDS, SHEAR ZONES, SEAMS, GAUGES, DYKES, AND FRACTURES. THESE DISCONTINUITIES TOGETHER WITH THE PLANES OF EXCAVATION CAN FORM A BLOCK WHICH IS UNSTABLE AND MAY FALL INTO THE OPENING.

DISCONTINUITY ANALYSIS METHOD





ANALYTICAL METHODS

DISCONTINUITY ANALYSIS METHOD...

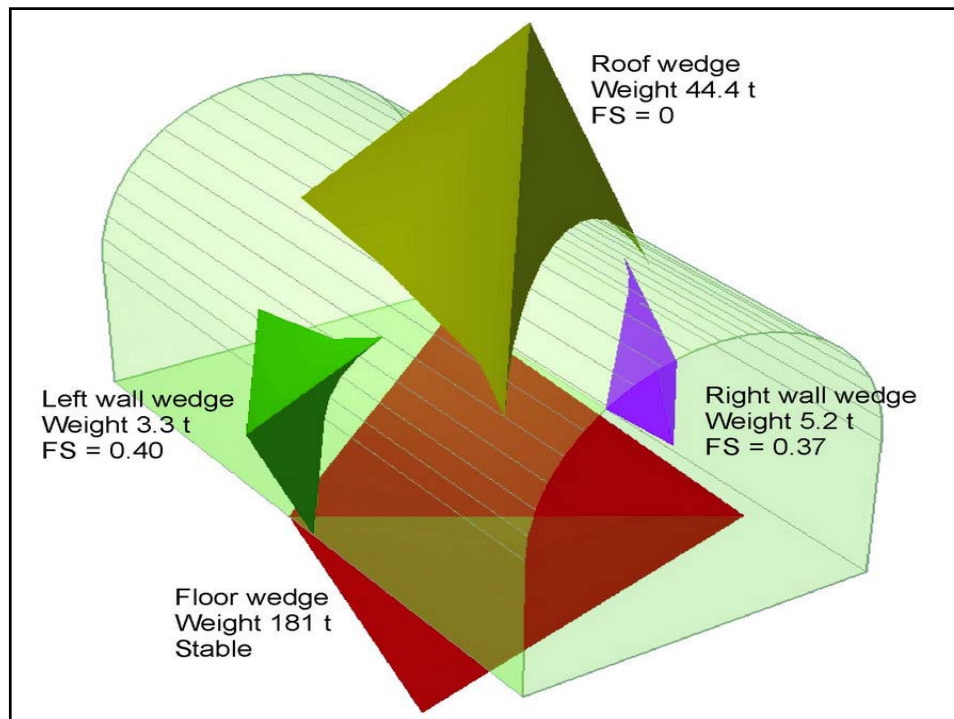
DEPENDING ON NUMBER OF JOINT SETS PRESENT AND THEIR ORIENTATIONS, ROCK MASS IS DIVIDED INTO INFINITE OR FINITE BLOCKS. FOR INFINITE BLOCKS, STABILITY IS NEVER A PROBLEM. IN FIGURE, THE FINITE BLOCKS MARKED 1, FORMED BY THE INTERSECTIONS OF AT LEAST THREE OR MORE SETS OF JOINTS, WILL FALL INTO THE OPENING IF FRICTION AT THEIR INTERFACES BECOME INCAPABLE OF RESISTING THE MOVEMENT. BLOCKS 2 WILL NOT SLIDE UNLESS BLOCKS 1 ALREADY HAVE FALLEN OUT. BLOCKS 3, 4 AND 5 WILL NOT FALL INTO THE OPENING UNLESS HYDROSTATIC PRESSURES MOVE THEM UPWARD.

ANALYTICAL METHODS

DISCONTINUITY ANALYSIS METHOD...

FOR ENSURING STABILITY, THE BLOCKS MUST REMAIN IN THEIR POSITION. THIS IS USUALLY DONE BY PROVIDING EXTERNAL SUPPORT.

GOODMAN (1988) INDICATED THAT IT IS ONLY NECESSARY TO PROVIDE STABILITY TO THE “KEY BLOCK.” HE DEFINED “KEY BLOCK” AS THE SINGLE BLOCK, THE REMOVAL OF WHICH WILL BRING UNCONTROLLABLE MOVEMENT OF SEVERAL OTHER BLOCKS THAT ARE SUPPORTED BY THE “KEY BLOCK”.



DISCONTINUITY ANALYSIS METHOD

For ensuring stability, they must remain in their position. This is usually done by providing external support. Once the weight of the movable block is determined, a rockbolt system with adequate factor of safety, usually 2, can be designed to ensure stability.

DISCONTINUITY ANALYSIS METHOD

It is only necessary to provide stability to the “key block.” “key block” defined as the single block, the removal of which will bring uncontrollable movement of several other blocks that are supported by the “key block”. Each block that has the potential of moving must be restrained by using rockbolts or any other external support system.

EMPIRICAL DESIGN RULES

- These rules, based on precedent practice, are available for determining design of rock bolt support system. These approaches provide a simple means by which the inexperienced designer can check for bolt length and spacing.
- Various empirical guide lines have been proposed by Coats & Cochrane (1970), the Canadian Ontario Hydro (1978), Hoek & Brown (1980), US Army Corps of Engineers (1980), Farmer & Shelton (1980), USBM (Lang & Bischoff 1982), Laubscher (1984), and others.

ANALYTICAL & NUMERICAL METHODS

Analytical methods involve the formulation and application of certain conceptual models for support design purpose.

Analysis of stresses and strains, of complicated geometrical shapes of openings, intersections of excavations with shafts or galleries, and complex geological environment require discretization of elements and materials. These analyses are very complex and more conducive to numerical methods than performing longhand calculations.

ANALYTICAL & NUMERICAL METHODS

- Two approaches to numerical modelling of rock masses can be identified, both recognizing geologic structures as being discontinuous due to joints, faults and bedding planes.
- A continuum approach treats the rock mass as a continuum intersected by a number of discontinuities, while a discontinuum approach views the rock mass as an assemblage of independent blocks or particles

ANALYTICAL & NUMERICAL METHODS

Continuum models are of two types:

1) Differential 2) integral

Differential models

It characterizes the entire region of interest and includes the **finite difference** and the **finite element methods**. The finite element method is uniquely capable of handling complex geometries and inhomogeneities. Complex systems involving fluid flow and heat transfer can also be handled.

ANALYTICAL & NUMERICAL METHODS

Integral or boundary element models

It feature discretization only along interior or exterior boundaries. The interfaces between different material types and discontinuities are treated as internal boundaries which must be similarly discretized. Boundary-element procedures are most appropriate for modelling linear-elastic systems, although certain forms of nonlinearity may be treated. The boundary-element procedures provide an economical means of two- and three-dimensional analysis of rock masses. They are particularly suitable for use when conditions at the boundary are of most concern.

ANALYTICAL & NUMERICAL METHODS

Discontinuum models

It feature numerical procedures involving the equations of motion of particles or blocks rather than the continuum. Discontinuum models should be used whenever independent rock block movements must be specifically recognized. This is the case of failure within a low stress field which allows large movements and rotations on the rock blocks defined by pre-existing geological discontinuities or by stress or blasting induced rock fracture.

ANALYTICAL & NUMERICAL METHODS

When assessing the potential or limitations of numerical techniques in underground opening, three areas of modelling activities should be considered:

- Model development;
- Model application;
- Model validation.

ANALYTICAL & NUMERICAL METHODS

In essence, for underground opening purposes, a wide range of two and three dimensional modelling capabilities is available with computer codes developed for many material behaviours.

However, success those models depends on the accuracy of the necessary input parameters which becomes very difficult in most of the cases.

A PRACTICAL EXAMPLE OF SUPPORT DESIGN USING Q-SYSTEM

Consider a drive of 9-m span in phyllite rock mass. The following is known:

Joint set 1: Smooth, planer $J_r = 1.0$

Chlorite coating $J_a = 4.0$

15 joints per meter

Joint set 2: Smooth, undulating $J_r = 2$

Slightly altered walls $J_a = 2$

5 joints per meter

Thus, $J_v = 15 + 5 = 20$ and $RQD = 115 - 3.3 J_v = 50\%$, $J_n = 4$

Minor water flows $J_w = 1.0$

Uniaxial compressive strength of phyllite rock $\sigma_c = 40$ MPa

Major principal stress $\sigma_1 = 3$ MPa

Minor principal stress $\sigma_3 = 1$ MPa

Hence, $\sigma_1 / \sigma_3 = 3$ and $\sigma_c / \sigma_1 = 13.3$ (medium stress), $SRF = 1.0$

$Q = 3.1$ (poor rock)

Support estimate: $B = 9$ m, $ESR = 1.6$

$B/ESR = 5.6$

For $Q = 3.1$: support category = 4

Support: untensioned rock bolts spaced 1.5 m, bolt length 2.5 m, and shotcrete 4-5 cm thick